

Nicholas Rhinehart

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Lab homepage: <https://leaf.utias.utoronto.ca> • Last modified: May 25, 2026

Current Position

University of Toronto, Assistant Professor Jul 2024–Present
Institute for Aerospace Studies • Department of Computer Science • Robotics Institute

Research focus: robot learning for safe, efficient autonomy, with emphasis on forecasting, planning, and control.

Education

Carnegie Mellon University, The Robotics Institute of the School of Computer Science Aug 2014–Sep 2019
Doctor of Philosophy in Robotics, Advisor: Kris Kitani

Carnegie Mellon University, School of Computer Science Jan 2013–Aug 2014
Master of Science in Robotics, Advisor: Drew Bagnell

Swarthmore College Aug 2008–May 2012
Bachelor of Arts in Computer Science, Bachelor of Science in Engineering

Professional Experience

University of Toronto, Institute for Aerospace Studies Jul 2024–Present
Assistant Professor • Toronto, ON, Canada

Waymo Research, Alphabet Nov 2022–Jun 2024
Senior Research Scientist • Mountain View, CA, USA

University of California, Berkeley, EECS and Berkeley AI Research (BAIR) Oct 2019–Sep 2022
Postdoctoral Scholar • Berkeley, CA, USA

University of California, Berkeley, EECS and Berkeley AI Research (BAIR) Jun 2018–Nov 2018
Visiting Researcher • Berkeley, CA, USA

NEC Labs America, Media Analytics Department May 2017–Sep 2017
Research Assistant • Cupertino, CA, USA

Uber Advanced Technologies Group Jun 2016–Sep 2016
Research Engineer • Pittsburgh, PA, USA

University of Tokyo, Institute of Industrial Science Jun 2015–Jul 2015
Visiting Researcher • Tokyo, Japan

Carnegie Mellon University, The Robotics Institute, School of Computer Science Aug 2014–Sep 2019
Doctoral Student Researcher • Pittsburgh, PA, USA

Carnegie Mellon University, The Robotics Institute, School of Computer Science Jan 2013–Aug 2014
Master's Student Researcher • Pittsburgh, PA, USA

Academic Awards

PAPER AWARDS

Best Poster Honorable Mention [6], Toronto Robotics Conference 2025

Best Paper Award [25], ICML 2019 Workshop on AI for Autonomous Driving 2019

Best Paper Award Honorable Mention [31], ICCV 2017 2017

FELLOWSHIP AWARDS

PhD Fellowship, CMU, Center for Machine Learning and Health 2018

IBM PhD Fellowship Finalist, CMU Robotics Institute 2017

The Robert E., Elizabeth, and Walter Lamb Scholarship, Swarthmore College 2011, 2012

PROFESSIONAL SERVICE AWARDS

Top Reviewer Award (x4), NeurIPS 2021 · ICML 2021 · NeurIPS 2019 · ICCV 2019 2019, 2019, 2021, 2021

Publications

PUBLICATION STATISTICS

Peer-reviewed publications ≥ 30 • **Best paper awards** ≥ 3 • **Total citations** ≥ 3000 • **h-index** ≥ 23 • **i10-index** ≥ 28

Publication statistics computed using Google Scholar.

MOST SIGNIFICANT CONTRIBUTIONS

- **Forecasting and planning in complex environments with deep generative models** [1, 2, 5, 8, 12, 21, 25, 30, 31]. I developed deep learning frameworks for robots to forecast the behaviors of people and plan their own actions jointly. Recent world-model foundations for planning and simulation include [1] and [2]. This direction started with [25] and [30], and later expanded to flexible planning with [21] and scene-forecasting results in [12]. This line of work includes forecasting papers used by follow-on work in robotics and autonomous driving, including [31], and has informed behavior-forecasting evaluation at Waymo [8]. Several papers in this line have accumulated substantial citations on Google Scholar, including [25]: ≥ 500 , [30]: ≥ 300 , [21]: ≥ 150 , [31]: ≥ 150 , and [8]: ≥ 100 .
- **Self-supervised generative world models for sensor forecasting** [1, 2, 7, 23]. A key technical thread in my work is learning generative models directly from raw sensor observations, without relying on manually annotated behavior labels. I showed that sequential pointcloud forecasting improves motion predictions by exploiting unlabeled sensor data [23], demonstrating that generative world models over raw observations can serve as implicit behavior models. This direction has scaled to self-supervised generative world models trained on large-scale occupancy data for realistic multi-agent traffic simulation [1] and long-horizon scene prediction [2], as well as generative radiance-field models for 3D scene forecasting [7]. [1] was evaluated on Waymo’s multi-agent simulation benchmark.
- **Intrinsic objectives for unsupervised control** [15, 18]. Inspired by the idea that biological systems do not have external reward signals analogous to those found in reinforcement-learning agents, I formalized notions of unsupervised control that do not rely on an external reward signal, and implemented them in a variety of settings [15] and [18]. In [15], by learning a policy to minimize average state entropy, agents learned structured behaviors, including playing Tetris and Doom and controlling a humanoid robot to avoid falls. In [18], related surprise-minimization objectives were shown to induce useful exploratory behaviors in dynamic environments.

PRE-PRINTS

- [1] [AutoWorld: Scaling Multi-Agent Traffic Simulation with Self-Supervised World Models](#)
M. Pourkeshavarz, T. Liu, **N. Rhinehart**
arXiv 2026. [Open Access](#).
- [2] [OccSim: Multi-kilometer Simulation with Long-horizon Occupancy World Models](#)
T. Liu, S. Zhao, M. Pourkeshavarz, W. Li, **N. Rhinehart**
arXiv 2026. [Open Access](#).
- [3] [Ratatouille: Imitation Learning Ingredients for Real-world Social Robot Navigation](#)
J. R. Han, H. Thomas, J. Zhang, **N. Rhinehart**, T. D. Barfoot
arXiv 2025. [Open Access](#).
- [4] [Residual Reward Models for Preference-based Reinforcement Learning](#)
C. Cao, M. Rogel-Garcia, M. Nabail, X. Wang, **N. Rhinehart**
arXiv 2025. [Open Access](#).

CONFERENCE AND JOURNAL PUBLICATIONS

- [5] [DR-MPC: Deep Residual Model Predictive Control for Real-world Social Navigation](#)
J. R. Han, H. Thomas, J. Zhang, **N. Rhinehart**, T. D. Barfoot
IEEE Robotics and Automation Letters (RA-L) 2025.
- [6] [Towards foundational LiDAR world models with efficient latent flow matching](#)
T. Liu, S. Zhao, **N. Rhinehart**
Neural Information Processing Systems (NeurIPS) 2025. [Open Access](#).

- [7] CARFF: Conditional Auto-encoded Radiance Field for 3D Scene Forecasting
J. Yang, K. Desai, C. Packer, H. Bhatia, **N. Rhinehart**, R. McAllister, J. Gonzalez
European Conference on Computer Vision (ECCV) 2024.
- [8] The Waymo Open Sim Agents Challenge
N. Montali, J. Lambert, P. Mouglin, A. Kuefler, **N. Rhinehart**, M. Li, C. Gulino, T. Emrich, Z. Yang, S. Whiteson, B. White, D. Anguelov
Neural Information Processing Systems (NeurIPS) 2023. [Open Access](#).
- [9] Hybrid Imitative Planning with Geometric and Predictive Costs in Off-road Environments
N. Dashora*, D. Shin*, D. Shah, H. Leopold, D. Fan, A. Agha-Mohammadi, **N. Rhinehart**, S. Levine
International Conference on Robotics and Automation (ICRA) 2022. [Open Access](#).
- [10] Is Anyone There? Learning a Planner Contingent on Perceptual Uncertainty
C. Packer, **N. Rhinehart**, R. McAllister, M. A. Wright, X. Wang, J. He, S. Levine, J. E. Gonzalez
Conference on Robot Learning (CoRL) 2022. [Open Access](#).
- [11] Offline Reinforcement Learning for Customizable Visual Navigation
D. Shah, A. Bhorkar, H. Leen, I. Kostrikov, **N. Rhinehart**, S. Levine
Conference on Robot Learning (CoRL) 2022. [Open Access](#).
- [12] S2Net: Stochastic Sequential Pointcloud Forecasting
X. Weng, J. Nan, K. Lee, R. McAllister, A. Gaidon, **N. Rhinehart**, K. Kitani
European Conference on Computer Vision (ECCV) 2022.
- [13] Conservative Safety Critics for Exploration
H. Bharadhwaj, A. Kumar, **N. Rhinehart**, S. Levine, F. Shkurti, A. Garg
International Conference on Learning Representations (ICLR) 2021. [Open Access](#).
- [14] Contingencies from Observations: Tractable Contingency Planning with Learned Behavior Models
N. Rhinehart*, J. He*, C. Packer, M. A. Wright, R. McAllister, J. E. Gonzalez, S. Levine
International Conference on Robotics and Automation (ICRA) 2021. [Open Access](#).
- [15] Information is Power: Intrinsic Control via Information Capture
N. Rhinehart, J. Wang, G. Berseth, J. D. Co-Reyes, D. Hafner, C. Finn, S. Levine
Neural Information Processing Systems (NeurIPS) 2021. [Open Access](#).
- [16] Parrot: Data-Driven Behavioral Priors for Reinforcement Learning
A. Singh*, H. Liu*, G. Zhou, A. Yu, **N. Rhinehart**, S. Levine
International Conference on Learning Representations (ICLR) 2021. [Open Access](#). Oral Presentation.
- [17] RECON: Rapid Exploration for Open-World Navigation with Latent Goal Models
D. Shah, B. Eysenbach, **N. Rhinehart**, S. Levine
Conference on Robot Learning (CoRL) 2021. Oral Presentation.
- [18] SMiRL: Surprise Minimizing RL in Dynamic Environments
G. Berseth, D. Geng, C. Devin, **N. Rhinehart**, C. Finn, D. Jayaraman, S. Levine
International Conference on Learning Representations (ICLR) 2021. [Open Access](#). Oral Presentation.
- [19] ViNG: Learning Open-World Navigation with Visual Goals
D. Shah, B. Eysenbach, G. Kahn, **N. Rhinehart**, S. Levine
International Conference on Robotics and Automation (ICRA) 2021. [Open Access](#).
- [20] Can Autonomous Vehicles Identify, Recover From, and Adapt to Distribution Shifts?
A. Filos*, P. Tigas*, R. McAllister, **N. Rhinehart**, S. Levine, Y. Gal
International Conference on Machine Learning (ICML) 2020. [Open Access](#).
- [21] Deep Imitative Models for Flexible Inference, Planning, and Control
N. Rhinehart, R. McAllister, S. Levine
International Conference on Learning Representations (ICLR) 2020. [Open Access](#).

- [22] Generative Hybrid Representations for Activity Forecasting with No-Regret Learning
J. Guan, Y. Yuan, K. M. Kitani, **N. Rhinehart**
Computer Vision and Pattern Recognition (CVPR) 2020. Oral Presentation.
- [23] Inverting the Forecasting Pipeline with SPF2: Sequential Pointcloud Forecasting for Sequential Pose Forecasting
X. Weng, J. Wang, S. Levine, K. Kitani, **N. Rhinehart**
Conference on Robot Learning (CoRL) 2020. Open Access.
- [24] Directed-Info GAIL: Learning Hierarchical Policies from Unsegmented Demonstrations using Directed Information
M. Sharma, A. Sharma, **N. Rhinehart**, K. M. Kitani
International Conference on Learning Representations (ICLR) 2019. Open Access.
- [25] PRECOG: PREDiction Conditioned On Goals in Visual Multi-Agent Settings
N. Rhinehart, R. McAllister, K. M. Kitani, S. Levine
IEEE International Conference on Computer Vision (ICCV) 2019. Open Access.
- [26] First-Person Activity Forecasting from Video with Online Inverse Reinforcement Learning
N. Rhinehart, K. M. Kitani
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2018.
- [27] Human-Interactive Subgoal Supervision for Efficient Inverse Reinforcement Learning
X. Pan, E. Ohn-Bar, **N. Rhinehart**, Y. Xu, Y. Shen, K. M. Kitani
International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2018. Open Access.
- [28] Learning Neural Parsers with Deterministic Differentiable Imitation Learning
T. Shankar, **N. Rhinehart**, K. Muelling, K. M. Kitani
Conference on Robot Learning (CoRL) 2018.
- [29] N2N Learning: Network to Network Compression via Policy Gradient Reinforcement Learning
A. Ashok, **N. Rhinehart**, F. Beainy, K. M. Kitani
International Conference on Learning Representations (ICLR) 2018. Open Access.
- [30] R2P2: A Reparameterized Pushforward Policy for Diverse, Precise Generative Path Forecasting
N. Rhinehart, K. M. Kitani, P. Vernaza
European Conference on Computer Vision (ECCV) 2018.
- [31] First-Person Activity Forecasting with Online Inverse Reinforcement Learning
N. Rhinehart, K. M. Kitani
IEEE International Conference on Computer Vision (ICCV) 2017. Oral Presentation; Best Paper Award Honorable Mention.
Awarded to 3 of 2,143 submissions.
- [32] Predictive-State Decoders: Encoding the Future into Recurrent Networks
A. Venkatraman*, **N. Rhinehart***, W. Sun, L. Pinto, M. Hebert, B. Boots, K. M. Kitani, J. A. Bagnell
Neural Information Processing Systems (NeurIPS) 2017. Open Access.
- [33] Learning Action Maps of Large Environments via First-Person Vision
N. Rhinehart, K. M. Kitani
Computer Vision and Pattern Recognition (CVPR) 2016.
- [34] Visual Chunking: A List Prediction Framework for Region-Based Object Detection
N. Rhinehart, J. Zhou, M. Hebert, J. A. Bagnell
International Conference on Robotics and Automation (ICRA) 2015.

PATENTS

- [35] Generative Adversarial Inverse Trajectory Optimization for Probabilistic Vehicle Forecasting
P. Vernaza, W. Choi, N. Rhinehart
US10739773B2, 2017.
- [36] Traffic prediction with reparameterized pushforward policy for autonomous vehicles
P. Vernaza, N. Rhinehart
US11189171B2, 2019.

[37] Balancing diversity and precision of generative models with complementary density estimators
P. Vernaza, N. Rhinehart, A. Liu, Kihyuk Sohn
US11049265B2, 2019.

Funding Awards

Funding source	% Alloc.	Role	Program Title	Proposal Title	Period / Status
University of Toronto	50%	P.I.	Joint Seed	Learning world models as data-driven simulators for robust robot manipulation of rigid and deformable objects	2026-09–2028-08 (upcoming)
NSERC	9%	Co-P.I.	CREATE	Agentic AI for Discovery in Health Sciences (AID4HS)	2026-09–2032-08 (upcoming)
NSERC	100%	P.I.	Discovery	Grounded World and Value Models for Reliable Model-Based Robot Autonomy	2026-04–2031-04 (active)
Digital Research Alliance of Canada	100%	P.I.	Resources for Research Groups	Learning and Distilling Unified World Models	2026-04–2027-04 (active)
LG Toronto	100%	P.I.		Advanced Robot Learning and Generalization	2025-11–2027-11 (active)
NVIDIA	100%	P.I.	Academic Grant Program	Learning Physically-Grounded World Models	2025-10–2026-04 (past)
Canada Foundation for Innovation (CFI) and Ontario Research Fund (ORF)	100%	P.I.	John R. Evans Leaders Fund (JELF)	Foundational World Models and Efficient Reward Learning for Robotics	2025-08–2027-03 (active)
Digital Research Alliance of Canada	100%	P.I.	Resources for Research Groups	Learning and Distilling Unified World Models	2025-07–2026-04 (past)
Google	100%	P.I.	TPU Research Cloud	Foundational World Models	2025-06–2026-06 (active)
University of Toronto, Institute for Aerospace Studies	100%	P.I.	Heuckroth Distinguished Faculty Award (lab startup funding)		2024-07– (active)
University of Toronto, Institute for Aerospace Studies	100%	P.I.	Student recruitment (lab startup funding)		2024-07– (active)
Toyota Research Institute	0%	Lead author (postdoctoral)	University 2.0	Universal Predictive Representations for Forecasting, Planning, and Control	2021–2024 (past)

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Funding source	% Alloc.	Role	Program Title	Proposal Title	Period / Status
Carnegie Mellon University, Center for Machine Learning and Health	100%	P.I.	Ph.D. Fellowship	Learning-Based Systems for Forecasting and Summarizing Human Behavior	2018-08–2019-08 (past)

Monetary values available upon request.

Academic and Professional Presentations

INVITED TALKS

<i>Learning What Matters for Model-Based Robotics</i> Toyota Research Institute , Los Altos, CA, USA [Remote]	May 2026
<i>Learning What Matters for Model-Based Robotics</i> University of Toronto, Robotics Institute Seminar Series , Toronto, ON, Canada	Apr 2026
<i>Key Ingredients of General-Purpose Robots</i> Toronto Robotics Conference , Toronto, ON, Canada	Jun 2025
<i>Contingency Planning with Learned Models of Behavioral and Perceptual Uncertainty</i> NeurIPS 2022 Workshop on Machine Learning for Autonomous Driving , New Orleans, LA, USA	Dec 2022
<i>Contingency Planning with Learned Models of Behavioral and Perceptual Uncertainty</i> IROS 2022 , Kyoto, Japan [Remote]	Oct 2022
<i>Developing Robots That Learn to Forecast and Affect the Future</i> University of Toronto Robotics Institute , Toronto, Canada	Jun 2022
<i>Developing Robots That Learn to Forecast and Affect the Future</i> Apple , Cupertino, CA, USA [Remote]	Jun 2022
<i>Developing Robots That Learn to Forecast and Affect the Future</i> McGill University, Department of Electrical Engineering , Digital	Mar 2022
<i>Developing Robots That Learn to Forecast and Affect the Future</i> Stanford University, Stanford Vision and Learning Lab , Digital	Jan 2022
<i>Developing Robots That Learn to Forecast and Affect the Future</i> U.C. Berkeley, Berkeley Artificial Intelligence Research Lab Seminar , Berkeley, CA, USA	Jan 2022
<i>Towards Learning to Forecast Everything for Making Complex Decisions</i> ICRA 2021 , Xi'an, China [Remote]	Jun 2021
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> Applied RL Seminar , Digital	Dec 2020
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> Uber ATG , Toronto, Canada [Remote]	Oct 2020
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> CVPR 2020 Workshop on Precognition , Seattle, WA, USA [Remote]	Jun 2020
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> Scale AI , San Francisco, CA, USA	Oct 2019
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> Tesla , Palo Alto, CA, USA	Oct 2019
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> ICCV 2019 Workshop on Autonomous Driving - Beyond Single Frame Prediction , Seoul, South Korea	Oct 2019
<i>Jointly Forecasting and Controlling Behavior from High-Dimensional Observations</i> Argo AI , Pittsburgh, PA, USA	Jul 2019
<i>Forecasting and Planning with PRECOG</i> iSee , Pittsburgh, PA, USA	May 2019

<i>Forecasting and Planning with PRECOG</i> Zoox , San Francisco, CA, USA	Jan 2019
<i>Deep Imitative Models for Flexible Inference, Planning, and Control</i> ACCV 2018 Attention/Intention Understanding Workshop , Perth, Australia	Dec 2018
<i>R2P2: A Reparameterized Pushforward Policy for Diverse, Precise Generative Path Forecasting</i> Waymo , Mountain View, CA, USA	Nov 2018
<i>R2P2: A Reparameterized Pushforward Policy for Diverse, Precise Generative Path Forecasting</i> U.C. Berkeley, Berkeley Deep Drive Group , Berkeley, CA, USA	Aug 2018
<i>Inverse Reinforcement Learning for Computer Vision</i> CVPR 2018 Tutorial on Inverse RL for Computer Vision , Organizer and Speaker, Salt Lake City, UT, USA	Jun 2018
<i>R2P2: A Reparameterized Pushforward Policy for Diverse, Precise Generative Path Forecasting</i> CVPR 2018 Tutorial on Human Activity Forecasting , Salt Lake City, UT, USA	Jun 2018
<i>R2P2: A Reparameterized Pushforward Policy for Diverse, Precise Generative Path Forecasting</i> U.C. Berkeley, RAIL Lab , Berkeley, CA, USA	Jun 2018
<i>R2P2: A Reparameterized Pushforward Policy for Diverse, Precise Generative Path Forecasting</i> NEC Labs America , Cupertino, CA, USA	Jun 2017
PANELS	
<i>Panelist</i>	
ICCV 2019 Workshop on Autonomous Driving - Beyond Single Frame Prediction , Seoul, South Korea	Oct 2019
CONTRIBUTED CONFERENCE AND WORKSHOP TALKS	
<i>Deep Imitative Models for Flexible Inference, Planning, and Control</i> BayLearn 2019 , San Francisco, CA, USA	Oct 2019
<i>Deep Imitative Models for Flexible Inference, Planning, and Control</i> NeurIPS 2018 Workshop on Infer2Control , Montreal, Canada	Dec 2018
<i>Deep Imitative Models for Flexible Inference, Planning, and Control</i> NeurIPS 2018 Workshop on ML for Intelligent Transportation Systems , Montreal, Canada	Dec 2018
<i>First-Person Activity Forecasting with Online Inverse Reinforcement Learning</i> ICCV 2017 , Venice, Italy	Oct 2017
<i>First-Person Activity Forecasting with Online Inverse Reinforcement Learning</i> MACV 2016 , Baltimore, MD, USA	May 2016
GUEST LECTURES	
Forecasting and Planning with PRECOG, CMU , Pittsburgh, PA, USA	May 2019
Deep Imitative Models for Flexible Inference, Planning, and Control, CMU , Pittsburgh, PA, USA	Nov 2018
Introduction to Computer Vision, Guest Lecture, CMU , Pittsburgh, PA, USA	Apr 2018
Graduate Statistical Techniques in Robotics, Guest Lecture, CMU , Pittsburgh, PA, USA	Apr 2018
Graduate Statistical Techniques in Robotics, Guest Lecture, CMU , Pittsburgh, PA, USA	Sep 2017

Academic Supervision

SUPERVISION STATISTICS

Active graduate supervision 7 • Past graduate supervision 1

GRADUATE STUDENT SUPERVISION

Yifan Ruan (UToronto PhD student, co-supervised)	2026-05–
Sebastian Aegidius (UToronto PhD student)	2025-09–
Chenyang Cao (UToronto PhD student)	2025-09–
Alex Pleava (UToronto MSc student)	2025-09–
Mozhgan Pourkeshavarz (UToronto PhD student)	2025-09–

Mohamed Nabail (UToronto PhD student)	2024-09–
Tianran Liu (UToronto PhD student)	2024-09–
James Han (UToronto MS student → Tesla, Humanoid Robotics)	2023-09–2025-08

Service and Professional Activities

CONFERENCE AREA CHAIR

CoRL '26	2026
ICML '26	2026
ICLR '25, '26	2025–2026
NeurIPS '23, '24, '25, '26	2023–2026

CONFERENCE AND JOURNAL REVIEWING

TMLR '22, '23, '24, '25, '26	2022–2026
ECCV '18, '20, '26	2018, 2020, 2026
JAIR '20, HRI '20, '25, TPAMI '20	2020, 2025
CoRL '20, '21, '22	2020–2022
ICML '19, '20, '21, '22	2019–2022
NeurIPS '19, '20, '21, '22	2019–2022
CVPR '16, '17, '18, '19, '22	2016, 2017, 2018, 2019, 2022
RA-L '21	2021
ICRA '19, '20, '21	2019–2021
ICCV '17, '19, '21	2017–2021
IJCV '19, TPAMI '19	2019

WORKSHOP ORGANIZER

ICRA '22 Workshop on Fresh Perspectives on the Future of Autonomous Driving	2022
NeurIPS '21 Workshop on Machine Learning for Autonomous Driving	2021
NeurIPS '20 Workshop on Machine Learning for Autonomous Driving	2020
ICML '19 Workshop on Imitation, Intent, and Interaction (I3)	2019
NeurIPS '19 Workshop on Machine Learning for Autonomous Driving	2019

TUTORIAL ORGANIZER

CVPR '18 Tutorial on Inverse RL for Computer Vision	2018
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GRANT REVIEWER

Digital Research Alliance of Canada Resource Allocation Competition	2026
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UNIVERSITY SERVICE

Graduate Department Appeals and Academic Committee (GDAAC) , Member, University of Toronto	2025–2026
Robotics Institute seminar series , Co-organizer, University of Toronto	2025–2026
UTIAS Planning Committee , Member, University of Toronto Institute for Aerospace Studies	2025–2026
aUToronto student group , Co-supervisor, University of Toronto	2025–2026
BAIR Undergraduate Mentoring , UC Berkeley	2020
Ph.D. Admissions Committee , CMU Robotics Institute	2017
Robotics Institute Representative , CMU Graduate Student Association	2015–2017
M.S. Admissions Committee , CMU Robotics Institute	2015, 2016
Class Treasurer , Swarthmore College	2012

Co-Chair, Swarthmore Philanthropy Council; assisted in coordinating alumni fundraising efforts 2011–2012

THESIS COMMITTEES AND EXTERNAL EXAMINATIONS

Gunshi Gupta, *external examiner*, Ph.D. student, University of Oxford 2026

Jasper Gerigk, *supervisory committee member*, Ph.D., UofT Computer Science 2026

Yimu Wang, *external examiner*, PhD student at University of Waterloo 2026

Jonah Phillion, *committee member*, Ph.D., UofT; Foundations and Applications of Trajectory Modeling for Autonomous Vehicles 2025

Arjun Sharma, M.S. Robotics, CMU; Integrating Structure with Deep Reinforcement and Imitation Learning 2018

Tanmay Shankar, M.S. Robotics, CMU; Learning Neural Parsers with Deterministic Differentiable Imitation Learning 2018

Teaching

INSTRUCTOR

Introduction to Learning from Data (ROB313), University of Toronto Spring 2026

Introduction to Robotics (ROB301), University of Toronto Fall 2025, Fall 2024

TEACHING ASSISTANT

Geometry-based Methods in Vision (16-822), Carnegie Mellon University Fall 2016

Data Structures and Algorithms (CPSC 035), Swarthmore College Fall 2011, Spring 2011

Introduction to Computer Science (CPSC 021), Swarthmore College Spring 2010

TUTOR

Fundamentals of Digital Systems (ENGR 015, CS 038), Swarthmore College Fall 2011